

Core Notation / Linear Maps

Linear map: $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ linear iff $f(u+v) = f(u) + f(v)$ and $f(\alpha u) = \alpha f(u)$; equivalently $f(\alpha u + \beta v) = \alpha f(u) + \beta f(v)$. Necessary quick check: $f(0) = 0$. Every linear map is uniquely $f(x) = Ax$ with columns $Ae_i = f(e_i)$. Matrix entries: A_{ij} is gain from input j to output i ; row i controls output i , column j is effect of input j .

Affine map: $f(\alpha u + \beta v) = \alpha f(u) + \beta f(v)$ for $\alpha + \beta = 1$ iff $f(x) = Ax + b$. Let $g(x) = f(x) - f(0)$; then g is linear and $b = f(0)$.

Linearization: for differentiable $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ near x_0 ,

$$f(x_0 + \delta x) \approx f(x_0) + Df(x_0)\delta x, \quad Df_{ij} = \frac{\partial f_i}{\partial x_j}(x_0).$$

Deviations obey $\delta y \approx Df(x_0)\delta x$.

Gradients: first order Taylor $f(z) \approx f(x) + \nabla f(x)^T(z - x)$. $\nabla(a^T x + b) = a$. $\nabla(x^T A x) = (A + A^T)x$; if $A = A^T$, $2Ax$.

Engineering Linear Models

Elastic structure: $y = Cx$, x_j force at node/direction j , y_i deflection i ; C_{ij} compliance in m/N. Column j is displacement pattern due to unit force j .

Static circuits: $y = Ax$ for independent source values x and circuit variables y ; if x currents and y voltages, A is impedance/resistance matrix. Dependent sources can still yield a linear map.

Mass with forces: unit mass, piecewise constant forces. Final velocity is integral of force; final position weights earlier forces more because they act longer. Force constraints plus desired final state $\Rightarrow$ linear equations/inequalities in force vector.

Thermal system: $y = Ax$, x_j heater powers, y_i steady-state temperature changes. A_{ij} is degrees C per watt from heater j to location i ; row i = heater effects at one location, column m = spatial heating pattern.

Interpretations

Composition/product view: AB means applying B first, then A . Column j of AB is A times column j of B ; row i of AB is row i of A acting on B . Entries of AB are inner products of rows of A with columns of B .

Mixture of columns: write $A = [a_1 \ a_2 \ \dots \ a_n]$. Then

$$Ax = x_1 a_1 + x_2 a_2 + \dots + x_n a_n.$$

Thus $y = Ax$ is a mixture of the columns of A , with mixture coefficients x_j . In control/design, column a_j is the response to unit input j , and x_j says how much of that actuator/response is used.

Inner product with rows: write the rows of A as $A = \begin{bmatrix} \tilde{a}_1^T & \dots & \tilde{a}_m^T \end{bmatrix}^T$. Then $y_i = \tilde{a}_i^T x$, $i = 1, \dots, m$.

Each output is an inner product of x with one row of A . In estimation, row i is sensor i 's sensitivity pattern; the measurement is large when x aligns with that row.

Useful identities: $(AB)^T = B^T A^T$, $(A^{-1})^T = (A^T)^{-1}$, $(A^T A)^T = A^T A$, $\text{tr}(AB) = \text{tr}(BA)$ when products make sense, $\text{diag}(x)u = x \odot u$.

Block/structured matrices: diagonal: outputs depend only on matching inputs. Lower triangular: output i depends only on inputs $1, \dots, i$. Permutations: $W = VP$ permutes columns of V ; $W = PV$ permutes rows.

Permutation matrices: each row/column has one 1. $P^{-1} = P^T$. They preserve norms, rank, independence, and orthogonality. Use them to reorder variables/columns without changing the underlying subspace.

Interpretations of $Ax = y$: estimation/inversion: y measurements, x unknown parameters, rows are sensors. Control/design: x inputs, y outcomes, columns are actuators/responses. Mapping/encoding: A transforms x to code/output y ; decoding asks whether preimage is unique.

Linear Algebra Review

Norms: $\|x\| \geq 0$, equality iff $x = 0$; $\|\alpha x\| = |\alpha| \|x\|$; $\|u+v\| \leq \|u\| + \|v\|$. Common norms: $\|x\|_2 = (x^T x)^{1/2}$, $\|x\|_1 = \sum_i |x_i|$, $\|x\|_p = (\sum_i |x_i|^p)^{1/p}$, $\|x\|_\infty = \max_i |x_i|$.

Unit balls: $B = \{x \mid \|x\| \leq 1\}$. In $\mathbb{R}^2$, the ℓ_1 ball is a diamond, the ℓ_2 ball is round, and the ℓ_∞ ball is a square.

Inner product: standard $\langle u, v \rangle = u^T v$; weighted $\langle u, v \rangle_W = u^T W v$ for $W = W^T \succ 0$. Properties: symmetry $\langle u, v \rangle = \langle v, u \rangle$; linearity $\langle \alpha u + \beta z, v \rangle = \alpha \langle u, v \rangle + \beta \langle z, v \rangle$; positive definite $\langle u, u \rangle \geq 0$, equality iff $u = 0$. Induced norm $\|u\| = \sqrt{\langle u, u \rangle}$. Orthogonal iff $u^T v = 0$. Pythagoras: if $u^T v = 0$, then $\|u+v\|^2 = \|u\|^2 + \|v\|^2$.

Cauchy-Schwarz / angles: $|u^T v| \leq \|u\|_2 \|v\|_2$, equality iff u, v linearly dependent. $\cos \theta = u^T v / (\|u\| \|v\|)$. Inner products give norms and angles; not every norm comes from an inner product (e.g. ℓ_1). For unit-norm embeddings, cosine similarity is $u^T v$.

Linear functionals / hyperplanes: a linear functional $f: \mathbb{R}^n \rightarrow \mathbb{R}$ has form $f(x) = c^T x$. Hyperplane $H = \{x \mid c^T x = d\}$ has normal c and dimension $n-1$ if $c \neq 0$. If $d = 0$, it passes through the origin; same normal $\Rightarrow$ parallel. Linear classifier: $\text{sign}(c^T x - d)$; normal points toward positive side.

Halfspaces / closeness: a halfspace is one side of a hyperplane: $H_- = \{x \mid c^T x \leq d\}$, $H_+ = \{x \mid c^T x \geq d\}$; the boundary is included in both closed halfspaces.

Points closer to a than b : $\|x - a\| \leq \|x - b\| \iff (b - a)^T x \leq (\|b\|^2 - \|a\|^2)/2$. Boundary is perpendicular bisector through $(a+b)/2$. Angle view: $c^T x < 0$ means obtuse ($90^\circ < \theta \leq 180^\circ$), $c^T x = 0$ means right ($\theta = 90^\circ$), and $c^T x > 0$ means acute ($0^\circ \leq \theta < 90^\circ$).

Subspaces / spans: $S \subseteq \mathbb{R}^n$ is a subspace iff $0 \in S$ and $\alpha u + \beta v \in S$ for all $u, v \in S$. Examples: $\mathbb{R}^n$, $\{0\}$, and $\text{span}(v_1, \dots, v_k)$. $\text{span}(v_1, \dots, v_k) = \{\alpha_1 v_1 + \dots + \alpha_k v_k\}$. Sum $U + V = \{u + v \mid u \in U, v \in V\}$ and intersection $U \cap V$ are subspaces. $U \cup V$ is a subspace iff $U \subseteq V$ or $V \subseteq U$. Orthogonal subspaces: $u^T v = 0$ for every $u \in U, v \in V$. Orthogonal complement $S^\perp = \{x \mid x^T s = 0 \ \forall s \in S\}$.

Independence: $v_1, \dots, v_k$ independent iff $\sum_i \alpha_i v_i = 0 \Rightarrow \alpha_i = 0$ for all i ; equivalently no v_i lies in the span of the others.

Basis: a basis is a nonredundant set of directions that generates the whole subspace: $v_1, \dots, v_d$ is a basis of S iff it spans S and is independent. Every $x \in S$ has a unique expansion $x = \sum_i \alpha_i v_i$; the coefficients α_i are its coordinates in that basis. Bases are not unique; coordinates depend on the chosen basis. Standard basis of $\mathbb{R}^n$: $e_1, \dots, e_n$. If $\dim S = d$, then d independent vectors in S form a basis; d spanning vectors are automatically independent.

Dimension: $\dim S = \#$ vectors in any basis. In a d -dimensional subspace, independent sets have at most d vectors and spanning sets have at least d .

Invertibility: square A invertible iff columns are independent iff columns form a basis of $\mathbb{R}^n$ iff $Ax = b$ has a unique solution for every b . A^{-1} undoes A : $A^{-1}Ax = x$ and $AA^{-1}w = w$. Proof snippets: if A invertible and $Ac = 0$, then $c = A^{-1}0 = 0$, so columns are independent. Conversely, if square A 's columns are independent, they form a basis of $\mathbb{R}^n$; solve $Ax = e_i$ for each i , stack solutions as B , then $AB = I$.

Change of coordinates / basis change: standard basis vectors are $e_1, \dots, e_n$, and $x = \sum_i x_i e_i$; the x_i 's are the coordinates of x in the standard basis. If $B = [b_1 \ \dots \ b_n]$ is another basis, $x = Bz = \sum_i z_i b_i$, and $z = B^{-1}x$ are the coordinates of x in the basis B . B maps B -coordinates to standard coordinates; B^{-1} maps standard coordinates to B -coordinates.

Similarity transformation: for linear transformation $y = Ax$, if $x = B\tilde{x}$ and $y = B\tilde{y}$, then $\tilde{y} = B^{-1}AB\tilde{x}$. $B^{-1}AB$ expresses the same linear transformation in B -coordinates. Similar matrices have the same rank and invertibility; $B^{-1}AB = I$ implies $A = I$.

Optimization geometry: minimizing distance to a subspace gives residual orthogonal to the subspace. If z is closest point to x in V , then $(x - z)^T v = 0$ for all $v \in V$.

Range and Nullspace

Nullspace: for $A \in \mathbb{R}^{m \times n}$ with rows r_i^T , $\text{null}(A) = \{x \in \mathbb{R}^n \mid Ax = 0\} = \{x \mid r_i^T x = 0, i = 1, \dots, m\}$. Thus $\text{null}(A)$ is orthogonal to every row. It is a set of directions/perturbations mapped to zero; also called the kernel.

Ambiguity from nullspace: if $y = Ax$ is a measurement of x , then $z \in \text{null}(A)$ is undetectable from the sensors: $A(x+z) = Ax = y$. Thus x and $x+z$ are indistinguishable from the measurement y ; $\text{null}(A)$ characterizes ambiguity in x given y . Control version: $z \in \text{null}(A)$ is an input with no result, so it gives freedom among inputs with the same output.

Zero nullspace / one-to-one / left inverse: for $A \in \mathbb{R}^{m \times n}$, these are equivalent: $\text{null}(A) = \{0\}$; A is one-to-one; columns are independent; there exists $L \in \mathbb{R}^{n \times m}$ with $LA = I_n$. Meaning: no nonzero input is crushed to zero, so the map loses no information.

Proof of one-to-one: $Ax_1 = Ax_2 \Rightarrow A(x_1 - x_2) = 0 \Rightarrow x_1 = x_2$; converse: $Az = 0 = A0 \Rightarrow z = 0$. Proof of left inverse: if $\text{null}(A) = \{0\}$, then $A^T A \in \mathbb{R}^{n \times n}$ is invertible since $A^T A c = 0 \Rightarrow c^T A^T A c = \|Ac\|^2 = 0 \Rightarrow Ac = 0 \Rightarrow c = 0$. Thus $L = (A^T A)^{-1} A^T$. Converse: $LA = I$, $Ax = 0 \Rightarrow x = LAx = 0$.

Also $\text{null}(A^T A) = \text{null}(A)$: if $c \in \text{null}(A)$, then $Ac = 0 \Rightarrow A^T A c = 0$, so $c \in \text{null}(A^T A)$; if $c \in \text{null}(A^T A)$, then $A^T A c = 0 \Rightarrow \|Ac\|^2 = 0 \Rightarrow Ac = 0$, so $c \in \text{null}(A)$. Left inverses need not be unique unless A is square invertible.

Range / consistency: for $A = [a_1 \ \dots \ a_n]$, $\text{range}(A) = \{Ax \mid x \in \mathbb{R}^n\} = \text{span}(a_1, \dots, a_n) \subseteq \mathbb{R}^m$. It is the set of vectors hit by the map $x \mapsto Ax$. $b \in \text{range}(A) \iff Ax = b$ is solvable. Estimation: possible/consistent sensor signals; $b \notin \text{range}(A)$ suggests sensor failure or model mismatch. Control/design: possible results or achievable outputs.

Onto / right inverse: for $A \in \mathbb{R}^{m \times n}$, onto means every desired output is hit: $\text{range}(A) = \mathbb{R}^m$, equivalently $Ax = b$ is solvable for every $b \in \mathbb{R}^m$. Equivalent conditions: columns span $\mathbb{R}^m$; rows are independent; $\text{null}(A^T) = \{0\}$; there exists $R \in \mathbb{R}^{n \times m}$ with $AR = I_m$. Onto $\Rightarrow$ right inverse: solve $Ax_i = e_i$ for each standard basis vector of $\mathbb{R}^m$, then $R = [x_1 \ \dots \ x_m]$ gives $AR = I_m$. Formula proof: if rows are independent, AA^T is invertible since $\text{null}(AA^T) = \text{null}(A^T)$, so $R = A^T(AA^T)^{-1}$ gives $AR = I_m$. Converse: if $AR = I_m$, then $x = Rb$ solves $Ax = b$ for every b . Control: choose $x = Rb_{\text{des}} \Rightarrow Ax = b_{\text{des}}$. Right inverses need not be unique unless A is square invertible. A right invertible iff A^T is left invertible.

Rank

Definition / basic facts: $\text{rank}(A) = \dim \text{range}(A)$. Nontrivial facts: $\text{rank}(A) = \text{rank}(A^T)$, and rank equals the maximum number of independent columns or rows. Hence $0 \leq \text{rank}(A) \leq \min(m, n)$. Column view: if a column is in the span of the others, remove it; repeat until the remaining columns are a basis for $\text{range}(A)$. **Conservation of dimension:** rank-nullity: $\text{rank}(A) + \dim \text{null}(A) = n$. Input dimensions are either hit in the output ($\text{rank}(A)$) or crushed to zero ($\dim \text{null}(A)$). Roughly: n input degrees of freedom = output degrees of freedom + lost degrees of freedom.

Coding interpretation: rank is the minimum code size needed to faithfully reconstruct $y = Ax$. If $A = BC$ with $C \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{m \times r}$, then $\text{rank}(A) \leq r$. Conversely, if $\text{rank}(A) = r$, then A factors through $\mathbb{R}^r$: encode x as $z = Cx \in \mathbb{R}^r$, decode $y = Bz$. The r numbers in z mediate every input-output pair.

Product rank facts: $\text{rank}(BC) \leq \text{rank}(B)$ since $\text{range}(BC) \subseteq \text{range}(B)$. Also $\text{rank}(BC) = \text{rank}((BC)^T) = \text{rank}(C^T B^T) \leq \text{rank}(C^T) = \text{rank}(C)$. Thus for $B \in \mathbb{R}^{m \times p}$, $C \in \mathbb{R}^{p \times n}$,

$$\begin{aligned} \text{rank}(B) + \text{rank}(C) - p &\leq \text{rank}(BC), \\ \text{rank}(BC) &\leq \min\{\text{rank}(B), \text{rank}(C)\}. \end{aligned}$$

This is Sylvester's inequality. Invertible pre/post multiplication preserves rank.

Low-rank computation / storage: store a rank- r factorization using $r(m+n)$ numbers instead of mn . Direct Ax costs about mn ; factored $B(Cx)$ costs about $rn+mr$, useful when $r \ll \min(m, n)$.

Full rank by shape: full rank means $\text{rank}(A) = \min(m, n)$. Skinny/tall $m \geq n$: columns independent, one-to-one, left invertible. Fat/wide $m \leq n$: rows independent, onto, right invertible. Square: full rank $\iff$ invertible $\iff$ one-to-one $\iff$ onto.

Rank dimensions / four spaces: with $r = \text{rank}(A)$:

$$\begin{array}{l|l|l} \text{range}(A) & \mathbb{R}^m & r \\ \text{null}(A^T) & \mathbb{R}^m & m-r \\ \text{range}(A^T) & \mathbb{R}^n & r \\ \text{null}(A) & \mathbb{R}^n & n-r \end{array}$$

$\text{range}(A)^\perp = \text{null}(A^T)$, $\text{range}(A^T)^\perp = \text{null}(A)$; $\mathbb{R}^m = \text{range}(A) \oplus \text{null}(A^T)$, $\mathbb{R}^n = \text{range}(A^T) \oplus \text{null}(A)$.

Useful identities: $\text{rank}([A \ B]) = \text{rank}(B) \iff \text{range}(A) \subseteq \text{range}(B)$. $\text{null}([A \ B]) = \text{null}(A) \cap \text{null}(B)$; $\text{rank}(\text{diag}(A, B)) = \text{rank}(A) + \text{rank}(B)$. If $M = N^T N$, then $x^T M x = \|Nx\|^2$ and $\text{null}(M) = \text{null}(N)$.

Orthogonality

Orthonormal set: normalized means $\|u_i\| = 1$; orthogonal means $u_i^T u_j = 0$ for $i \neq j$; orthonormal means both, so $u_i^T u_j = \delta_{ij}$. For $U = [u_1 \ \dots \ u_k]$, orthonormal columns mean $U^T U = I_k$. Orthonormality is a property of a set, not one vector. Orthonormal sets are independent: multiply $\sum_i \alpha_i u_i = 0$ by u_j^T to get $\alpha_j = 0$. They form an orthonormal basis for $\text{span}(u_1, \dots, u_k) = \text{range}(U)$.

Orthogonal matrices / warning: square U is orthogonal if $U^T U = I$; then also $U U^T = I$, and columns/rows are orthonormal bases for $\mathbb{R}^n$. Thus $U^{-1} = U^T$. If $U \in \mathbb{R}^{m \times k}$ is tall with $U^T U = I_k$, do not assume $U U^T = I_m$; $U U^T$ projects onto $\text{range}(U)$ and has rank at most k .

Expansion / coordinates: if square U is orthogonal, then $U^T x$ gives components in the u_i directions and

$$x = U(U^T x) = \sum_i (u_i^T x) u_i,$$

$$\|x\|^2 = \sum_i (u_i^T x)^2 \quad (\text{Parseval}).$$

No inverse is needed; U^T resolves x into components and U reconstitutes it. In an orthonormal basis, coordinates are inner products. Energy is preserved, and large coefficients reveal dominant directions; drop small ones for compression/denoising. Common bases: Fourier/DCT, wavelets, PCA.

Complementary subspaces: if $U = [U_1 \ U_2]$ is orthogonal, then $\text{range}(U_1)$ and $\text{range}(U_2)$ are complementary orthogonal subspaces. They are orthogonal: every vector in one is orthogonal to every vector in the other. Every $x \in \mathbb{R}^n$ is a sum of one vector from each subspace:

$$x = U_1 U_1^T x + U_2 U_2^T x.$$

Each subspace is the orthogonal complement of the other: $\text{range}(U_1)^\perp = \text{range}(U_2)$ and $\text{range}(U_2)^\perp = \text{range}(U_1)$. Proof idea: $U_1^T U_2 = 0$; if $w \perp \text{range}(U_1)$, then $U_1^T w = 0$, so $w = U_2 U_2^T w \in \text{range}(U_2)$. For any subspaces $S, T \subseteq \mathbb{R}^n$,

$$\begin{aligned} \dim(S + T) &= \dim S + \dim T - \dim(S \cap T), \\ (S + T)^\perp &= S^\perp \cap T^\perp, \quad (S \cap T)^\perp = S^\perp + T^\perp. \end{aligned}$$

The decomposition $x = s + t$ is unique iff $S \cap T = \{0\}$, written $S \oplus T$.

Geometry preservation: if $U^T U = I$, then $(Ua)^T (Ub) = a^T b$, $\|Ua\| = \|a\|$, and distances/angles are preserved. Square orthogonal maps are rotations or reflections.

Projection principle: the closest point $z \in V$ to x is characterized by $x - z \in V^\perp$. Projection onto nonzero q : $\text{proj}_q x = (q^T x / q^T q) q$, or $(q^T x) q$ if $\|q\| = 1$. For full column rank A , projection onto $\text{range}(A)$ is $P = A(A^T A)^{-1} A^T$. An orthogonal projector satisfies $P = P^T = P^2$, $\text{range}(I - P) = \text{range}(P)^\perp = \text{null}(P)$, and $x = Px + (I - P)x$ orthogonally. Hence $\|Px\| \leq \|x\|$, with equality iff $x \in \text{range}(P)$, while $Px = 0$ iff $x \in \text{range}(P)^\perp$. If P, Q are orthogonal projectors, PQ is an orthogonal projector iff $PQ = QP$; then it projects onto $\text{range}(P) \cap \text{range}(Q)$.

QR Factorization

Gram-Schmidt procedure: start with independent $a_1, \dots, a_n$. G-S constructs orthonormal q_i 's with the same progressive spans: $\text{span}(q_1, \dots, q_k) = \text{span}(a_1, \dots, a_k)$. At step j , remove the part of a_j already in previous q_i directions, then normalize:

$$\tilde{q}_j = a_j - \sum_{i < j} (q_i^T a_j) q_i, \quad q_j = \tilde{q}_j / \|\tilde{q}_j\|.$$

If the a_i 's are independent, $\tilde{q}_j \neq 0$ at every step. **Matrix form:** with $A = [a_1 \cdots a_n]$, $Q = [q_1 \cdots q_n]$,

$$A = QR, \quad Q \in \mathbb{R}^{m \times n}, \quad Q^T Q = I, \quad R \in \mathbb{R}^{n \times n}$$

where R is upper triangular and invertible. Entries satisfy $r_{ij} = q_i^T a_j$ for $i < j$, $r_{jj} = \|\tilde{q}_j\|$, and $r_{ij} = 0$ for $i > j$. Thus $a_j = r_{1j} q_1 + \cdots + r_{jj} q_j$. Columns of Q are an orthonormal basis for $\text{range}(A)$. Usually QR is computed with modified G-S or Householder, which is less sensitive to rounding than classical G-S.

General G-S / staircase: if the columns are dependent, some $\tilde{q}_j = 0$; then a_j is a linear combination of previously generated q_i 's, so skip it. On exit, $Q \in \mathbb{R}^{m \times r}$, where $r = \text{rank}(A)$, is an orthonormal basis for $\text{range}(A)$, and $A = QR$ with R in upper staircase form. Nonzero corner entries are exactly the columns that generated a new q_i ; their count is the rank. One G-S run also gives QR factorizations for all leading column blocks.

Applications in lecture: QR directly gives an orthonormal basis for $\text{range}(A)$ and shows which columns were dependent on earlier ones. To remove a direction, $x_\perp = x - (q^T x) q = (I - qq^T)x$; with subspace basis Q , the removed component is $QQ^T x$ and the remainder is $(I - QQ^T)x$. G-S depends on column order. Replacing $(a_1, a_2, \dots)$ by $(a_1, a_1 + a_2, \dots, a_1 + \cdots + a_n)$ preserves every leading span, so Q is unchanged up to signs, but R changes.

Full QR: if $A = Q_1 R$ is thin QR, extend Q_1 to a full orthogonal basis $Q = [Q_1 \ Q_2]$ by applying general G-S to $[Q_1 \ B]$ where $[Q_1 \ B]$ has full rank. Then

$$A = [Q_1 \ Q_2] \begin{bmatrix} R \\ 0 \end{bmatrix}, \quad \text{range}(Q_2) = \text{range}(Q_1)^\perp.$$

This is the lecture statement that any orthonormal set can be extended to an orthonormal basis for $\mathbb{R}^m$.

Pivoted QR: a permutation matrix P can move useful columns to the front:

$$AP = QR.$$

The pivoted columns are the ones that generate new q_i 's. This makes the staircase pattern/rank easier to read and is rank-revealing.

Orthogonal decomposition induced by A : apply full

QR to A^T . If $A \in \mathbb{R}^{m \times n}$ and $A^T = [Q_1 \ Q_2] \begin{bmatrix} R \\ 0 \end{bmatrix}$, then $\text{range}(Q_1) = \text{range}(A^T)$. Also $AQ_2 = 0$, so $\text{range}(Q_2) = \text{null}(A)$. Hence

$$\mathbb{R}^n = \text{range}(A^T) \oplus \text{null}(A), \quad x = Q_1 Q_1^T x + Q_2 Q_2^T x$$

uniquely. This gives $\text{null}(A) = \text{range}(A^T)^\perp$, $\dim \text{range}(A^T) + \dim \text{null}(A) = n$, and $\text{rank}(A) = \text{rank}(A^T)$.

Projection/solve via QR: if $A = QR$ full column rank, projection onto $\text{range}(A)$ is QQ^T . Exact $Ax = b$ becomes $QRx = b$; if consistent, $Rx = Q^T b$. Least-squares later also solves $Rx = Q^T b$ without forming $A^T A$.

Linear Equations / Consistency Tricks

Solving $Ax = b$: exact solution exists iff $b \in \text{range}(A)$ iff $\text{rank}(A) = \text{rank}([A \ b])$. If A full column rank: at most one solution. If A full row rank: at least one solution for every b . General solution when consistent:

$$x = x_p + z, \quad z \in \text{null}(A).$$

Solution count: no solution if inconsistent. Unique solution iff consistent and $\text{null}(A) = \{0\}$. Infinitely many iff consistent and $\dim \text{null}(A) > 0$. For wide $m < n$, if consistent then infinitely many.

Rank tests to memorize: columns independent $\Leftrightarrow \text{rank}(A) = n$. Rows independent $\Leftrightarrow \text{rank}(A) = m$. Columns span $\mathbb{R}^m \Leftrightarrow \text{rank}(A) = m$. Rows span $\mathbb{R}^n \Leftrightarrow \text{rank}(A) = n$.

Homogeneous systems: $Ax = 0$ always has $x = 0$. Nonzero solution iff columns dependent iff $\text{rank}(A) < n$. Wide $m < n$ always has nontrivial nullspace.

Independence: columns independent iff $Ax = 0 \Rightarrow x = 0$. Nonnegative independence: if every $u_i \neq 0$ and $u_i^T u_j \geq 0$ for all i, j , then $\sum_i \alpha_i u_i = 0$, $\alpha_i \geq 0$ implies all $\alpha_i = 0$ because norm squared is a sum of nonnegative terms.

Orthogonal ranges: if $A^T B = 0$ and A, B are skinny full rank, then $[A \ B]$ is full rank: $Au + Bv = 0 \Rightarrow \|Au\|^2 + \|Bv\|^2 = 0$.

Rank-one matrix: uv^T has rank 1 if $u, v \neq 0$. Trace $\text{tr}(uv^T) = u^T v$. Column space is $\text{span}(u)$; row space is $\text{span}(v)$.

PSD/null trick: if $M = N^T N$, then $x^T M x = \|Nx\|^2$ and $\text{null}(M) = \text{null}(N)$. For sums $M = N_1^T N_1 + N_2^T N_2$, $x^T M x = \|N_1 x\|^2 + \|N_2 x\|^2$.

Augmented systems: if A is $m \times n$, $[A \ b]$ adds one column. Consistency is unchanged by elementary row operations. If inconsistent, least-squares later solves closest point in $\text{range}(A)$.

How to prove subspace: show 0 is in the set, then closure under $\alpha u + \beta v$. For sets defined by homogeneous linear equations, e.g. $Ax = 0$, this is immediate. Nonhomogeneous sets $Ax = b$, $b \neq 0$, are affine, not subspaces.

HW1 Patterns

Testing linearity from probes: check $f(0) = 0$, additivity $f(u + v) = f(u) + f(v)$, homogeneity $f(\alpha u) = \alpha f(u)$. To identify a linear $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, probe e_1, e_2 ; matrix columns are $f(e_1), f(e_2)$.

Signal-flow graphs: entry A_{ij} equals sum over all directed paths from input j to output i of product of edge gains. Cascaded identical systems: if $u = Ax$ and $z = Au$, then $z = A^2 x$.

Path-count proof: $(A^2)_{ij} = \sum_k A_{ik} A_{kj}$ counts length-2 walks through intermediate node k . Inductively, $(A^k)_{ij}$ counts length- k walks. For undirected graphs $A = A^T$.

Matrix language examples:

- Columns of C are combos of columns of B : $C = BF$.
- Rows of Z from rows of Y : $Z = FY$ with row-combination matrix F .
- Column p_i acute with q_j : $P^T Q$ has positive entries; corresponding columns acute: $\text{diag}(P^T Q) > 0$.
- First k columns of A orthogonal to rest: if $A = [A_1 \ A_2]$, then $A_1^T A_2 = 0$.

Graph powers: for undirected adjacency A , $(A^k)_{ij}$ counts walks of length k from node i to node j .

Signal processing matrices: upsample/interpolate/downsample are linear; each output row contains the coefficients of the relevant samples. Downsample $y_i = x_{2i}$: row i has a 1 in column $2i$. Average pairs: row i has 1/2 in columns $2i - 1, 2i$.

2x interpolation pattern: $y_{2k-1} = x_k$ and $y_{2k} = (x_k + x_{k+1})/2$. Matrix has alternating rows: standard basis rows, then adjacent averaging rows.

Affine proof pattern: $f(x) = Ax + b$ is affine because $\alpha + \beta = 1$ preserves offset. Converse: $g(x) = f(x) - f(0)$; show $g(\alpha u + \beta v) = \alpha g(u) + \beta g(v)$ and set $A = [g(e_1) \cdots g(e_n)]$.

HW1 gradients: for scalar f , gradient is column vector satisfying $df \approx \nabla f^T dx$. For $f = x^T A x$, use scalar expansion:

$$df = (dx)^T A x + x^T A dx = x^T (A^T + A) dx.$$

So $\nabla f = (A + A^T)x$.

HW2 Patterns

Cauchy-Schwarz proof trick: $\|v + \lambda w\|^2 = v^T v + 2(v^T w)\lambda + w^T w \lambda^2 \geq 0$. Nonnegative quadratic $\forall \lambda$ gives $|v^T w| \leq \sqrt{v^T v} \sqrt{w^T w}$.

Right inverse example method: solve $Az = y$ symbolically; use free variables to enforce row/triangular constraints. If $AB = I$ then B has trivial nullspace: $Bx = 0 \Rightarrow x = ABx = 0$. A zero column of B would force zero column of AB , impossible.

HW2 P5 template: for

$$A = \begin{bmatrix} -1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix},$$

all right inverses have

$$B = \begin{bmatrix} s_1 & s_2 & s_3 \\ t_1 & t_2 & t_3 \\ -t_1 & 1 - t_2 & -t_3 \\ -s_1 & -s_2 & 1 - s_3 \\ 1 & 0 & 1 \end{bmatrix}.$$

Feasible cases: (a) row 2 zero, (d) rows 2 and 3 equal, (f) lower triangular. Impossible: nullity one, zero third column, upper triangular.

True/false traps: nonnegative entries + onto is not preserved by addition. $\text{null}([A; A + B; A + B + C]) = \text{null}(A) \cap \text{null}(B) \cap \text{null}(C)$. $\text{null}([A; AB; ABC])$ is not generally intersection of three nullspaces.

More TF traps: If $A^T A$ onto, then columns of A independent, but A need not be onto when tall. If $\begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}$

full rank, both blocks full rank. If A^2 onto, A onto.

HW2 TF answer pattern: (a) false via $I + \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$; (b) true by subtracting equations; (c) false with scalars;

(d) true by PSD/null trick; (e)(f)(g) true; (h) false with tall vector; (i)(j) true.

Wireless SINR: with $D = \text{diag}(G_{ii})$, $K = G - D$,

$$Dp = S(\sigma 1 + Kp) \Rightarrow (D - SK)p = S\sigma 1.$$

Assuming invertible, candidate $p = (D - SK)^{-1} S\sigma 1$; feasible iff $0 \leq p_i \leq P^{\max}$. For HW2 data largest $S \in \{2, \dots, 4\}$ is 3.8 with $p \approx (0.0948, 0.0474, 0.0547)$.

SINR derivation detail: equation per receiver:

$$G_{ii} p_i = S \left(\sigma + \sum_{j \neq i} G_{ij} p_j \right).$$

No optimization needed: for fixed S the full-rank assumption gives one candidate p , then just check bounds.

Rational interpolation:

$$f(x) = \frac{a_0 + a_1 x + \cdots + a_m x^m}{1 + b_1 x + \cdots + b_m x^m}.$$

For each datum:

$$a_0 + \cdots + a_m x_i^m - y_i b_1 x_i - \cdots - y_i b_m x_i^m = y_i.$$

Linear in unknowns $(a_0, \dots, a_m, b_1, \dots, b_m)$. Increase m until consistent. HW2 data: $m = 5$.

Interpolation matrix row: row i for degree m is

$$[1 \quad x_i \quad \cdots \quad x_i^m \quad -y_i x_i \quad \cdots \quad -y_i x_i^m].$$

Check residual $\|A\theta - y\|_\infty$ and final interpolation error $\max_i |f(x_i) - y_i|$.

Rational caveat: after solving, denominator $1 + b_1 x + \cdots + b_m x^m$ must be nonzero at sample points; otherwise original equation is not valid as a finite function value.

HW3 Patterns

Orthogonal complement facts: $V^\perp$ is a subspace. If $V = \text{span}(v_i) = \text{range}(V)$, then $V^\perp = \text{null}(V^T)$. Every v decomposes uniquely as $v + v^\perp$.

Unique decomposition proof: if $v_1 + w_1 = v_2 + w_2$ with $v_i \in V$, $w_i \in V^\perp$, then $v_1 - v_2 = w_2 - w_1 \in V \cap V^\perp$. Its norm squared is zero, so both parts are equal.

Robot coin collector: $y(t) = t$. With piecewise constant forces f_j on $[2j - 2, 2j)$ and mass 1,

$$x(t) = \sum_{j=1}^n a_{tj} f_j$$

$$a_{tj} = \begin{cases} 0, & t \leq 2j - 2 \\ \frac{1}{2}(t - 2j + 2)^2, & 2j - 2 < t < 2j \\ 2(t - 2j + 1), & t \geq 2j. \end{cases}$$

Collecting specified coins is linear system $Af = x_{\text{coin}}$. All coins possible iff consistent. All but one: delete each row and test consistency. HW3: miss coin 7; $f = (1, -4, 7, -10, 20, -35)$.

Robot coefficients intuition: if force interval is already finished at time t , contribution is velocity impulse $2f_j$ times remaining travel plus within-interval displacement, giving $2(t - 2j + 1)f_j$.

Layered medium: choose $x_1 = 1$. Unknowns $x_2, \dots, x_n, y_1, \dots, y_n$. Equations:

$$x_{i+1} = t_i x_i + r_i y_{i+1}, \quad y_i = r_i x_i + t_i y_{i+1}, \quad y_n = x_n.$$

Solve linear system; $S = y_1/x_1 = y_1$. Nominal $n = 20$, $t = .96$, $r = .02$: $S \approx 0.49818$. Fault scan: set one $(t_k, r_k) = (.02, .96)$, compare $|S_k - S_{\text{fault}}|$; for $S_{\text{fault}} = 0.70$, best $k = 9$.

Layered system setup: unknown vector can be ordered as $(x_2, \dots, x_n, y_1, \dots, y_n)$. There are $2(n - 1)$ interface equations plus final equation $y_n = x_n$, total $2n - 1$ equations.

Fault location logic: measured S is one scalar, but finite candidate faults make brute force possible: compute S_k for each interface and choose smallest absolute error. Last interface excluded because it is already fully reflective.

Practice Midterm Fast Checks

Answer patterns: implication questions: prove by definitions or give small counterexample. Common answers from practice: $A \Leftrightarrow B$, neither, $A \Rightarrow B$. For true/false, test 0, I , $2I$, rank-one uv^T , tall/skinny, wide/fat, and two lines in $\mathbb{R}^2$.

Fast traps: $A^T A$ is PSD and $A^T A = 0 \Rightarrow A = 0$; if $A^T A$ is upper triangular, symmetry makes it diagonal, so columns are orthogonal, not necessarily unit. Wide $A \in \mathbb{R}^{m \times n}$, $m < n$: either no solution or infinitely many.

$\text{range}(A) \subseteq \text{range}(B) \iff \text{rank}([A \ B]) = \text{rank}(B)$. Infinitely many solutions imply $\text{null}(A) \neq \{0\}$; converse needs consistency.

Orthogonal/rank checks: if $U^T U = I$, then $U U^T$ projects onto $\text{range}(U)$, so $\{x \mid U U^T x = x\} = \text{range}(U)$. For $A = uv^T$, all nonzero columns are scalar multiples of u ; G-S returns $\pm u/\|u\|$. $\text{rank}(FG) \leq \text{rank}(F)$, $\text{rank}(G)$; full-rank square factors preserve rank. A nonzero nilpotent can have $F^2 = 0$.

Counterexamples: $S \cup T$ is a subspace only when one subspace contains the other; two distinct lines through the origin disprove it. Halfspace not span: lacks negatives/origin. Null product trap: Bx can land in $\text{null}(A)$. $A^T A$ diagonal but not orthogonal: $A = 2I$. Nonnegative

onto sum can fail: $I + \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has rank 1.